

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE/IT)

May 2012

## IT205 OPERATING SYSTEM

Date: 17.05.2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

### Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

### SECTION - I

- Q - 1** Consider the following set of processes in order P1, P2, P3, P4 and P5 with the length [07]  
of the CPU burst time given in milliseconds. Calculate average waiting time and turn-around time using given scheduling algorithm. Draw Gantt Chart using following Scheduling algorithm. (1) SJF (Pre-emptive) (2) RR (quantum=2)

| Process | Arrival Time | Burst Time |
|---------|--------------|------------|
| P1      | 1            | 10         |
| P2      | 4            | 9          |
| P3      | 3            | 5          |
| P4      | 0            | 7          |
| P5      | 2            | 4          |

- Q - 2 (a)** What is Deadlock? Explain mechanisms to recover from the deadlock. [05]  
(b) A disk has 1000 cylinders, numbered 0 to 999. The drive is currently serving a request [05]  
at cylinder 135, and the queue of pending requests is:

89, 470, 913, 775, 948, 509, 33, 11, 134, 230

Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for SSTF and C-SCAN disk-scheduling algorithms?

- (c) Explain file system layout stored on the disk. [04]

### OR

- Q - 2 (a)** Consider the following snapshot of a system: [05]

|    | Allocation |   |   |   | Max |   |   |   | Available |   |   |   |
|----|------------|---|---|---|-----|---|---|---|-----------|---|---|---|
|    | A          | B | C | D | A   | B | C | D | A         | B | C | D |
| P0 | 0          | 0 | 1 | 2 | 0   | 0 | 1 | 2 | 1         | 5 | 2 | 0 |
| P1 | 1          | 0 | 0 | 0 | 1   | 7 | 5 | 0 |           |   |   |   |
| P2 | 1          | 3 | 5 | 4 | 2   | 3 | 5 | 6 |           |   |   |   |
| P3 | 0          | 6 | 3 | 2 | 0   | 6 | 5 | 2 |           |   |   |   |
| P4 | 0          | 0 | 1 | 4 | 0   | 6 | 5 | 6 |           |   |   |   |

Using Banker's algorithm, Is the system in a safe state? If yes, show safe sequence.

- (b) Explain the operation of a DMA transfer. [05]  
(c) What is i-node? Explain its pros and cons. [04]



Q - 3 (a) Consider the following page reference string: [06]

6, 4, 2, 4, 3, 1, 7, 2, 1, 3, 2, 1, 4, 5, 6

How many page faults would occur for LRU and Optimal page replacement algorithms, assuming four frames? Initially pages 7, 5, 3 and 1 are loaded in the main memory.

(b) For a created process, the memory manager allocates free memory using various algorithms. Explain any three algorithms. [06]

(c) Average process size is 16384 bytes, page size is 1 Kbytes, and each page entry contains 6 bytes. Calculate (a) approximate number of pages needed per process and (b) total overhead due to page table and internal fragmentation loss. [02]

OR

Q - 3 (a) With reference to FIFO page replacement algorithm, explain Belady's anomaly for available page frames 3 and 4 using example. [06]

(b) Explain the need and working of multiple page tables in detail. [06]

(c) A virtual address is 010001000100. MMU converts this address into 11-bit physical address. Page size is 256 bytes. Calculate number of pages and page frames? [02]

### SECTION - II

Q - 4 (a) Explain terms: (i) Race condition (ii) Demand paging. [03]

(b) Differentiate: Monolithic system - Microkernel [04]

Q - 5 (a) Explain three state process transitions with diagram. [05]

(b) Explain cat, grep and chmod commands of Unix with example. [05]

(c) Explain the different mechanisms to secure the data? [04]

OR

Q - 5 (a) What is mutual exclusion? How can it be achieved? [05]

(b) Write a shell script to that calculates sum of a digit of a given number. [05]

(c) Differentiate: Starvation - Deadlock. [04]

Q - 6 (a) Explain thread and multi threading environment with example. [07]

(b) Explain the impact of the impact of global allocation policy on paging system compare to local policy. [07]

OR

Q - 6 (a) Explain Producer-consumer problem and its solution using semaphores. [07]

(b) Discuss the Dining Philosophers problem with reference to processes. [07]

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